



Society of Petroleum Engineers

SPE-229213-MS

Turning Risk into Quantifiable Asset. Platform-Based Project Execution Powered by Data Science and AI

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229213-MS](https://doi.org/10.2118/229213-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

The accelerating energy transition presents a dual challenge for developers: managing a growing portfolio of renewable projects across diverse geographies while delivering large-scale, capital-intensive infrastructure to exacting standards. This complexity is compounded by fragmented data, siloed processes, and conventional project management tools that fail to provide timely, actionable insights, resulting in persistent schedule delays, cost overruns, and underperforming assets.

This paper demonstrates how a three-pronged digital approach, combining ontology-based semantic lakes for digital continuity, context-aware intelligence and analytics, and collaborative platform-based virtual twin experiences, transforms project risk from a liability into a strategic asset. A semantic lake integrates and contextualizes data across enterprise systems, creating an ontology-driven knowledge graph that supports advanced queries, traceability, and adaptive integration of new data sources. Role-based data perspectives and virtual twin experiences allow stakeholders to visualize project status in real time, detect early warning signals, and take coordinated action with full situational awareness.

Case studies from Larsen & Toubro Energy-Hydrocarbon (LTEH) and EDF illustrate the tangible benefits of this approach. LTEH achieved early risk identification, improved coordination across EPC activities, and realized measurable gains in efficiency, schedule adherence, and cost control. EDF leveraged digital continuity and AI to unify requirements management, enhance system interface alignment, and industrialize nuclear project execution, improving predictability across multiple concurrent programs.

AI and ML extend these capabilities by enabling predictive risk forecasting, schedule optimization, knowledge-driven issue resolution, generative requirements and traceability, sourcing optimization, and virtual companions. These tools extract patterns from historical data, generate actionable insights, automate repetitive tasks, and embed recommendations within the virtual twin environment. Across use cases, organizations gain faster decision-making, improved collaboration, and higher execution reliability.

Collectively, the methodology demonstrates that integrating digital continuity, collaborative platforms, and AI-driven intelligence enables energy developers to anticipate challenges, optimize resources, and manage projects proactively. By turning risk into foresight, complexity into control, and uncertainty into a

lever for competitive advantage, this approach represents a transformative model for next-generation project delivery in the energy sector

Introduction

As the energy transition accelerates, energy companies face a dual challenge: managing a growing portfolio of medium-to-small renewable energy projects spread across diverse geographies, while simultaneously delivering large-scale, capital-intensive energy infrastructure and facilities to exacting standards. On one hand, global expansion demands standardized processes; on the other, it must allow for local autonomy to meet varying regulations and market conditions. At the same time, large infrastructure projects require seamless orchestration of thousands of interconnected processes, where even minor missteps can trigger costly delays and overruns.

Today, energy projects are more financially exposed than ever. Early risk identification and response are not optional—they are essential. Yet, conventional project management and risk tools remain inadequate. Even with sophisticated planning tools, teams often struggle to understand how a single late shipment, design change, or missing component will ripple through schedules, budgets, and quality. Too often risks go undetected, downstream impacts are misunderstood, and corrective actions come too late. The result: persistent cost overruns, schedule slippages, and underperforming assets.

Recent advances in digital technologies now make it possible to transform risk from a liability into a manageable, quantifiable, even strategic asset.

The current project delivery landscape of digital tools and approaches reveals 3 critical gaps.

1. Speed, Accuracy and Consistency.

Most digital solutions lack the ability to deliver reliable, real-time assessments of project health. Teams rely on incomplete or conflicting data, delaying recognition of risks and limiting effective response.

2. Coordinated and Aligned Team Actions.

Effective risk mitigation requires well-coordinated, cross-functional team collaboration. Today's siloed systems often hinder this, limiting the ability to share knowledge, lessons learned, and best practices across projects and regions.

3. Predictive and Actionable Insights

Organizations struggle to detect "weak signals" of risk that foreshadow serious issues and turn them into timely actionable insights. There is a pressing need for predictive capabilities that improve forecasting accuracy for schedules, budgets, and resource planning, enabling more proactive and less corrective execution

This paper explores a novice setup and application of 3 digital technologies to closing these gaps:

1. **Semantic Lake for Digital Continuity.** An ontology-based semantic lake provides an intelligent digital backbone that ingests, interprets, and contextualizes data from diverse enterprise systems, including PLM, ERP, MES, IoT, and simulation tools. By organizing information into knowledge graphs, it reveals relationships across project data, supports advanced queries and graph navigation, enables adaptive integration of new sources, and ensures full traceability. This structure transforms fragmented and siloed data into a unified, living representation of project status.
2. **Context-Aware Intelligence and Analytics.** Building on the semantic lake, data perspectives are created for stakeholders through virtual twin experiences on a collaborative platform. Structured data, unstructured text, and historical records are analyzed using AI and machine learning pipelines to deliver predictive, diagnostic, and prescriptive insights. Data Scientist and Data Linguist roles

work in concert to uncover risks, detect weak signals, automate analytics, and generate actionable recommendations tailored to each user's context.

3. **Collaborative Platform and Virtual Twin Experiences.** Dynamic virtual twins integrate role-based data perspectives, live 3D models, and real-time project metrics to provide a single, interactive environment for planning, monitoring, and decisionmaking. The collaborative platform connects internal teams, contractors, and suppliers, enabling transparent communication, task assignment, KPI tracking, and scenario simulations. Teams can explore "what-if" scenarios, visualize dependencies, and coordinate corrective actions without leaving the platform, ensuring alignment across geographies and disciplines.

Together, these components and three-pronged digital approach create a coherent digital ecosystem that combines semantic intelligence, predictive analytics, and immersive collaboration. It turns raw data into actionable knowledge, supports proactive risk management, and enables real-time decision-making.

The result is a unified, platform-based digital environment, adaptable to local needs, that detects risks at the earliest and enables instant action. Enhanced with machine learning and predictive analytics, organizations learn from past projects to identify patterns, anticipate challenges, and optimize resource allocation. Teams can act proactively rather than reactively, accelerating decision-making and execution.

We will examine case study results of two companies, Larsen & Toubro (LTTE) and EDF, that have adopted this approach. They have implemented the methodology and core fundamentals of the technology components, while collaborating closely with Dassault Systemes. The result is a true partnership model that is shaping the next advancements in project execution with a key focus on risk identification, quantification and response applying Data Science, Machine Learning and AI. This will be covered through practical use cases in real-world settings, demonstrating tangible business value.

Approach & Setup

This section outlines the approach taken, detailing the digital capabilities, apps and tools used, the digital environment that was set up and the collaborative processes deployed to create a step-change in energy project execution.

Digital Continuity - The Reality for Energy Developers

Digital continuity is only as powerful as the technology that underpins it, requiring an intelligent digital backbone that can ingest, interpret, and contextualize information across disciplines. For energy developers, turning siloed and fragmented data into meaningful, real-time insights remains a core objective and at the same time a core challenge. The first is connecting and making accessible data from enterprise systems such as PLM, ERP, MES, IoT, and simulation tools. The second is extracting actionable insights without overwhelming the teams.

Traditional tools and approaches often fall short because their data connection processes, such as APIs and ETLs, are fragile, expensive, and difficult to scale. Even when data is consolidated, most solutions merely aggregate data without enriching or contextualizing it, leaving raw information devoid of the semantics, relationships, and dependencies required for meaningful insight.

Additional limitations arise when updates lag behind, rendering the "single source of truth" outdated in fast-moving projects. As data volumes expand from IoT, BIM, engineering and supplier sources, many systems struggle with scalability, performance or the flexibility to account for local autonomy or regulatory differences, prompting users to build "shadow systems" that recreate the very silos the digital continuity was meant to eliminate.

Semantic lake approach applies a new generation of digital capabilities that addresses and better manages these shortcomings. It applies ontologies, domain-specific semantic structures, that organize information into knowledge graphs. These reveal relationships across the data, transforming complexity into a shared

understanding of "what's happening" and "what needs to happen next." The approach that was applied in this paper two case studies enables expansive queries (graph traversal, semantic, and join in parallel), adaptive integration of new sources without remodeling, and even the creation of new ontologies via generative AI. Together, these capabilities establish a living, execution-driven view of projects that can evolve with changing demands.

What sets this approach apart is its ability to blend advanced search, analytics, and semantic graph technologies with 3D-aware features. This means data can be analyzed directly on models, while full-text search, natural language processing, faceting, and graph traversal allow deep exploration of interconnected structures, configurations, and lifecycles. Built with AI governance and traceability, the platform is well-suited for regulated industries and can even support domain-specific language models, such as a materials ontology.

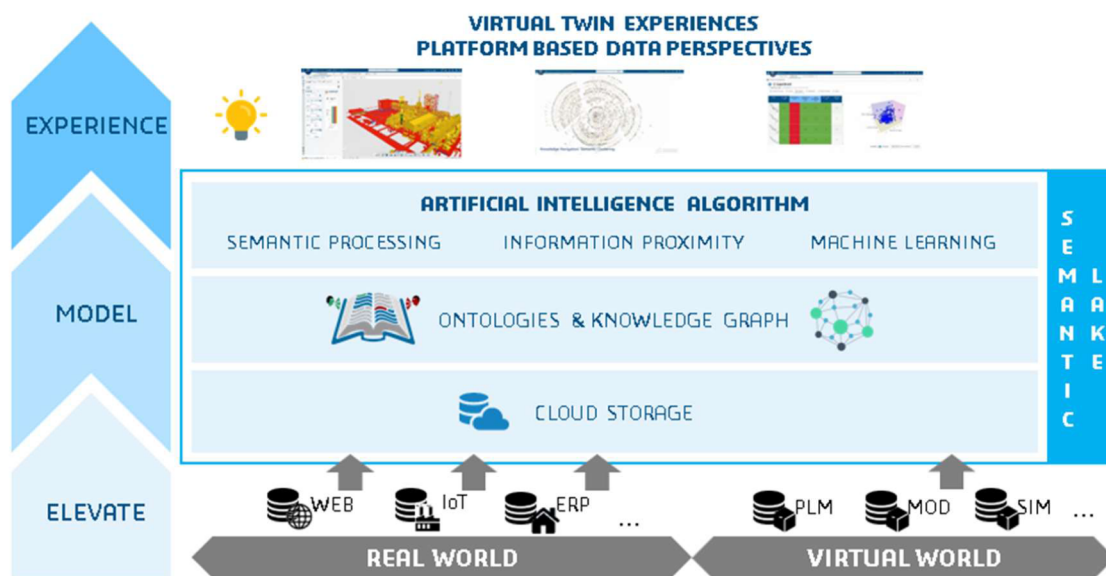


Figure 1—The semantic lake approach explained

Behind this environment is a data and AI ecosystem that ensures information flows seamlessly from raw input to business impact. Five core data functions and capabilities, Data Steward, Data Engineer, Data Scientist, Data Linguist, and Data Analyst, work in concert to create the semantic lake and power the AI and Data Science ecosystem.

and surveys. By combining AI-driven techniques with human expertise, it identifies topics, themes, risks, and emerging signals within large text collections. This ensures textual insights are aligned with enterprise ontologies and structured data, unlocking untapped value, improving knowledge discovery, and strengthening strategic decision-making.

The Data Scientist capability transforms structured data into actionable insights by designing and executing AI pipelines and machine learning models. This involves preparing and transforming datasets, training and evaluating models using methods such as classification and regression, and selecting optimal configurations for deployment. Once models are published, they are configured for inference or retraining and monitored throughout their lifecycle to ensure accuracy and efficiency. Through these activities, the Data Scientist delivers predictive and prescriptive insights, automates decision-making, and scales AI adoption across the enterprise.

Virtual Twin Experiences on a collaborative platform

Virtual twin experiences provide dynamic digital replicas of energy projects, assets, and infrastructure, designed for direct human interaction with live data in the semantic lake. They integrate role-based data perspectives and update automatically as design changes, supply chain updates, and cost data flows into the semantic lake, giving stakeholders a familiar 3D context to answer critical questions: Where are the delays? What's causing them? Which issues are urgent?, without searching across systems or doing manual analysis.

The Data Analyst capabilities transform datasets into business-ready insights.

They explore and query the data and knowledge graphs, apply filters and aggregations, and create visualizations such as tables, charts, and maps. It bridges business needs with data-driven answers with interactive dashboards that present tailored perspectives for stakeholders, ensuring insights are accessible, understandable, and actionable. Role-based data perspectives can be deployed by default, while individuals can adapt or create their own analyses, simulations, and "what-if" scenarios.

The Data Viewer extends this value by enabling project members to navigate shared perspectives, interact with KPIs, and collaborate on analytics within the virtual twin. By directly engaging with the common referential of the virtual twin, teams can reveal intelligence from data, identify levers for optimization, and take coordinated action on performance indicators.

The collaborative platform unifies people, data, and processes, balancing global standards with local flexibility. Teams use live dashboards, embedded workflows, formal and informal means of communication and task management to assign responsibilities, monitor dependencies, and resolve issues quickly. Social communities foster open dialogue and knowledge sharing across internal teams, suppliers, contractors, and consultants, surfacing problems early and enabling coordinated responses.

For energy developers, these capabilities are transformative. Key data streams defined at project initiation support real-time oversight of supplier performance, material flow, and schedule adherence, allowing leaders to detect risks early, trace root causes, and validate readiness, all in a single, integrated environment. Developed in close collaboration with customers, the solution continuously evolves through real-world feedback, refining predictive models and actionable insights.

Cloud deployment ensures global accessibility, scalability, and security, while integrated analytics and simulation transform execution into a connected, data-driven process that drives faster, smarter, and more reliable outcomes.

Case Studies

Two case studies illustrate how leading energy developers are applying digital continuity, collaborative platforms, and advanced analytics to improve project execution results. The examples highlight practical implementations in complex, multi-stakeholder environments, showing how organizations connect

engineering, procurement, and construction data to enable early risk detection, predictive insights, and coordinated decision-making across the project lifecycle.

LTEH achieves early risk identification and control

Larsen & Toubro Energy-Hydrocarbon (LTEH) is a leader in delivering, complex energy facilities to the highest quality, but also to increasingly aggressive budgets and schedules. In projects of this scale, even the smallest hitch in the engineering, procurement and construction (EPC) process can cause major delays and budget overspend.

From design to engineering, procurement and even paying suppliers, it's crucial to get things right first time, manage changes and ensure quality every step of the way. They fundamentally re-imagined how to harvest the data from across their project execution systems and empower teams with actionable insights, risk identification capabilities and the environment to plan and execute effectively

LTEH's transformation began with the adoption of a collaborative platform environment to establish digital continuity, integrated schedules, issue management, and KPI-based dashboards to give stakeholders a consistent, real-time view of progress and performance.

This evolved into immersive VR reviews for early issue detection, and a "multi-D twin" linking 3D CAD models with schedules, budgets, safety, quality, and collaboration metrics, creating a live, execution-driven view of projects

Shifting from Visibility to Prediction. Embedded analytics enable teams to run "what-if" scenarios, quantify impacts, and coordinate corrective actions in the same environment where execution happens. Supply chain performance is continuously monitored, with automated alerts guiding targeted interventions.

By connecting core data and processes, LTEH now detects risks before they escalate. Using the semantic lake, they analyze engineering, procurement, and quality metrics to spot subtle "weak signals," generating predictive alerts earlier than traditional monitoring. Generative AI is next, further reducing risk by optimizing plans, resources, and embedding lessons learned.

From an initial push for digital continuity to today's advanced project intelligence, LTEH has turned complex, multi-stakeholder projects into coordinated success stories.

A practical example of early risk detection and learning from recent project history is the following use case: using completed tasks to predict the end dates of tasks still in progress. A BERT-type algorithm converts task descriptions into vectors, allowing the system to measure similarity between tasks. Based on these similarities, each ongoing task is given an estimated end date derived from the most comparable completed tasks.

EDF is driving digital transformation in nuclear PROJECTS

EDF, the world's largest producer of carbon-free electricity, is entering a new era of nuclear development, shifting from bespoke, one-off projects to an industrialized model delivering multiple EPR reactors, mid-sized EPR1200s, and the new NUWARD™ SMR in parallel. Managing this unprecedented scale demanded more than engineering excellence, it required a fundamental shift in how EDF manages and leverages data across the lifecycle of its nuclear programs.

From Digital Backbone to Engineering Intelligence. EDF launched its Digital Continuity program on a collaborative platform to unify engineering processes. Today, the platform connects hundreds of users, managing millions of components, systems, and materials for EPR design, creating a robust foundation for data integrity. Building on this, EDF deployed data science capabilities to transform raw data into actionable intelligence, targeting one of nuclear engineering's most complex areas: systems engineering and requirements management.

This new environment delivers:

1. *Unified Requirements Management*: A single, cross-disciplinary view with full traceability, linking requirements to product architecture and components for more informed engineering decisions.
2. *Change and Risk Control* : Continuous monitoring of requirement volatility, validation of modifications, and early risk detection to maintain maturity and reduce costly issues.
3. *System Interface Alignment* : Early identification and resolution of system-to-system misalignments to prevent downstream rework.

With these capabilities, EDF engineers can visualize, validate, and act on interconnected data faster and more accurately, strengthening compliance, detecting issues earlier, and improving predictability across multiple concurrent programs, paving the way for a repeatable, industrialized approach to nuclear construction.

Results

In our case studies, the semantic lake set-up, creating data perspectives consumable in virtual twin experiences on a collaborative platform, integrates about 80% of company system data. This shift enables these organizations to move beyond reporting on outcomes to actively shaping them, helping teams manage complexity, improve visibility, and deliver with precision.

The virtual twin experiences on a collaborative platform methodology adopted by these early adopters are yielding improvements in several key areas of project execution, realizing up to 25% gains in efficiency and 10-15% reductions in project durations. L&T's deployment showed real-time collaboration gains of 5-10%, reduced total installed costs by 2-5%, and streamlined EPC operations across complex programs. EDF on their side has seen a decrease by up to 90% of non-conformances and expects close to 100% traceable requirements coverage.

Here is a selection of examples of topics and capabilities that have seen a step-change in performance:

Proactive Risk & Issue Management. Teams perform 360-degree diagnostics across programs, projects and tasks using visualizations like S-curves, Gantt charts, and variance graphs to assess health in real time. Risks are detected early, prioritized through analytics, and linked directly to related technical data. Critical issues are highlighted for investigation within the virtual twin, alternatives can be tested, root causes are uncovered early with supporting data in context for faster, more collaborative resolution.

Cross-Project & Resource Intelligence. Portfolio-wide visibility makes it easy to spot workload imbalances and optimize staffing plans across geographies and disciplines. Overloaded team members are identified quickly, and tasks can be reassigned in real time via an interactive Kanban view—boosting utilization, eliminating bottlenecks, and maintaining productivity.

Scenario Planning & Schedule Optimization. Schedules can be managed precisely using baselines and "what-if" scenarios that test alternatives without disrupting the master plan. Options are compared against baselines to assess their impact on cost, delivery, and resources, with approved changes pushed seamlessly to execution. Updates made in the 3D environment ensure stakeholders see the latest plan and understand its broader implications.

Deliverable Tracking in 3D Context. Deliverables are tracked for gate reviews with full traceability back to project requirements and activities. Delays can be evaluated in 3D, with cross-highlighting between models and schedules to clarify impacts for all disciplines. This visual context ensures changes and risks are consistently understood, communicated, and acted upon.

Global Collaboration and Knowledge Sharing. By breaking down silos, the platform connects disciplines, organizations, and regions, enabling local expertise to feed into global strategies and drive alignment.

Automate task assignments, track dependencies, and manage formal reviews, while integrating with enterprise systems for phased, scalable adoption of digital practices

The value is clear: the semantic lake approach enhances transparency, accountability, and coordination across all stakeholders and all phases of project delivery. Supply chain alignment becomes more resilient, collaboration more seamless, and governance more robust. In short, digital continuity powered by AI and semantic intelligence turns project risk and complexity into a strategic advantage.

Ongoing Development and Future Work With A Focus On Machine Learning And AI

Machine Learning (ML) and Artificial Intelligence (AI) are emerging as integral components of the virtual project twin, extending its capabilities in cost and schedule optimization, resource allocation, and proactive risk identification. As AI adoption accelerates across industries, its application in complex project execution is becoming a strategic focus.

In this context, the ongoing development of AI-powered methods is being carried out in close collaboration with our industry partners, including with the companies in our 2 case studies in this paper. It ensures that new solutions and development directly address operational challenges encountered in practice.

These early implementations and ongoing research topics already demonstrate tangible impact across multiple fronts: from faster issue resolution and more accurate project risk forecasting to automated requirements generation, optimized sourcing and design, better knowledge reuse, and the introduction of AI-powered virtual companions. Together, these advances highlight how AI is moving beyond experimentation to deliver measurable value in real project execution.

The capabilities of these AI-enabled digital tools can be broadly characterized into four categories: (A) predictive capabilities, (B) generative and automation approaches, (C) optimization methods, and (D) virtual companions. Collectively, these directions not only address current limitations in project execution but also define a trajectory for the future of digitally enabled project delivery and management.

The following selection of topics, currently under deployment, testing or development with LTEH and EDF, illustrates how ongoing research is being translated into practical, AI-enabled solutions for project execution.



Figure 3—Overview of AI use case deployed, tested or under development

AI for Project Risk Forecasting (Deployed) and Schedule Optimization (Testing)

Energy infrastructure projects face chronic risks of delay and cost overruns due to unforeseen bottlenecks, shifting scopes, and inefficient resource use. AI mitigates these risks by forecasting likely deviations in schedule and cost under current execution conditions and recommending proactive adjustments. Building on this, schedule optimization capabilities predict task durations, identify risks, and recommend adjustments before execution begins. By integrating schedule monitoring with real-time execution data, AI enables project leaders to align master plans with field conditions, simulate alternatives, and reduce late-stage surprises. This leads to more predictable delivery and higher success rates in complex projects.

AI for knowledge driven issue resolution (Testing)

Recurring problems in project execution and maintenance, such as construction delays or equipment failures, often arise not from lack of expertise but from fragmented data and reliance on "tribal knowledge." AI addresses this by creating an intelligent, centralized knowledge base that captures alerts, failure patterns, root causes, and best-practice resolutions. Natural Language Processing (NLP) extracts insights from documents, emails, and historical records, building a structured, searchable repository embedded into digital twins and workflows.

When new issues occur, the system rapidly identifies similar past cases and provides context-aware recommendations, accelerating response times, standardizing actions, and reducing costly downtime. Teams, including new engineers, gain immediate access to institutional knowledge, improving onboarding, strengthening cross-functional collaboration, and empowering more confident, informed decision-making throughout the project lifecycle.

AI for Generative Requirements and Traceability (Under Development)

Managing stakeholder requirements in engineering projects has traditionally been a tedious, error-prone process, involving the conversion of scattered Word documents and spreadsheets into formal, structured, and traceable specifications. In many projects, this step is skipped altogether—leading to downstream issues such as sub-optimal system design, costly errors, and missed requirements that prevent the asset or infrastructure from meeting AI transforms this by automatically generating precise, standards-aligned requirements directly linked to design models, verification criteria, and test cases. This reduces ambiguity, accelerates validation, and ensures compliance. The Generative Requirements Engineer exemplifies this approach, resolving interdependencies, shortening certification cycles, and enabling systematic traceability between requirements and design. The outcome is higher-quality specifications that adapt quickly to evolving standards, turning requirements management into a lever for both agility and assurance.

AI for Sourcing Optimization (deployed in other industries), Part Design, and Additive Manufacturing (testing in other industries)

Managing large inventories often leads to redundancy, excess stock, and inflated procurement costs. AI addresses this by analyzing 3D shapes, models and metadata to identify duplicate or similar parts, consolidate sourcing, and maximize reuse across assets. Building on this foundation, AI-driven generative product design allows engineers to explore countless design variants within defined parameters, optimizing geometry for weight, strength, manufacturability, and cost. Virtual validation reduces prototypes, accelerates development, and supports mass customization, helping teams innovate while maintaining efficiency.

For additive manufacturing, AI combines 3D model metadata with enterprise spend data to rapidly assess feasibility and cost-effectiveness across thousands of parts. Material intelligence further ensures compliance and sustainability by flagging high-risk or regulated substances, supporting responsible sourcing and regulatory adherence.

Together, these methods cut procurement costs, accelerate development cycles, and support compliance with materials regulations, turning sourcing and design into competitive differentiators.

AI-powered virtual companions (testing)

AI-powered assistants in the collaborative platform boost productivity, simplify complex tasks, and make knowledge instantly accessible through natural language interaction. Acting as a bridge between users and the platform's vast ecosystem of data, services, and applications, they transform how teams work across engineering, manufacturing, program management, and more.

Using large language models (LLMs) and advanced AI, these assistants understand intent, retrieve relevant knowledge from Dassault Systèmes' industrial data and customer-specific content, and provide input to take meaningful action. They can answer questions, find documentation, execute design tasks, generate reports, and automate processes, reducing manual effort, errors, and search time.

With secure connections to enterprise data, responses are traceable, explainable, and compliant with company policies. Contextual awareness allows them to learn from interactions and deliver increasingly relevant support. Typical use cases include creating parts, modifying digital twins, retrieving supplier data, and updating project statuses.

By integrating knowledge discovery, task execution, and process automation in one intelligent interface, these assistants accelerate decision-making, promote knowledge reuse, and improve collaboration, especially in large, distributed organizations. Built on a scalable AI infrastructure, they evolve with new functions and integrate seamlessly into existing workflows, driving agility, efficiency, and faster time-to-market.

How It Works (Across All Use Cases)

Across these applications, AI operates through a consistent set of mechanisms applying the semantic lake capabilities such as data scientist described earlier:

- *Pattern Recognition:* Models are trained on historical project data, technical documentation, and operational records to identify anomalies, predict risks, and surface insights.
- *Real-Time Contextual Analysis:* Live data streams from sensors, schedules, supply chains, and operations are continuously compared against learned benchmarks to generate timely alerts and recommendations.
- *Generative and Automation Techniques:* AI translates unstructured inputs into structured outputs (e.g., requirements, design variants), automating tasks traditionally handled manually.
- *Integration with the Virtual Twin:* Knowledge, forecasts, and recommendations are embedded within the Virtual Twin environment, ensuring cross-functional teams have a common understanding and access to knowledge.

This shared foundation ensures that each deployment is not a standalone tool but part of a coherent digital ecosystem that scales across projects and domains

Conclusion: Turning Risk Into A Strategic Asset

In the modern energy sector, risk isn't just a variable to watch, it's a resource to harness. By uniting digital continuity, robust platform execution, and AI/ML-driven intelligence on the 3DEXPERIENCE platform, risk becomes measurable, predictable, and actionable.

Predictive models and real-time analytics uncover hidden patterns and provide early warning signals before they escalate. This empowers teams to simulate scenarios, test alternatives, and prepare for multiple outcomes with confidence.

With data-driven insights, project managers can optimize resources, align stakeholders, and act decisively to keep execution on track. Delays, cost overruns, and operational constraints shift from being reactive challenges to proactively managed opportunities.

This is more than risk mitigation, it's risk transformation. With the right data foundation and intelligent tools, energy developers turn uncertainty into foresight, complexity into control, and risk into a lever for competitive advantage.

Dassault Systèmes partners with customers to push the technology frontier, applying digital technologies and implementation methodologies to maximize business impact. Real-world successes with LTEH & EDF and a growing portfolio of AI use cases demonstrate both our innovation and our track record. The opportunity now is to scale these capabilities across the enterprise, turning uncertainty into a lasting competitive advantage.

The question is no longer whether to transform, but how quickly you can seize the advantage.